

Neva Krien

AI developer and technical educator with experience turning AI research into practical demos, course material, and developer-focused content. Strong background in Python, Rust, LLM workflows, and low-level systems programming.

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Experience

AI Research & Education

- Mentee of Guy Tamir (Intel) since 2022; contributed to research and educational outreach in AI.
- Participated in a research team focused on large language models (LLMs) for C++ code generation. See the team's [first paper](#), published August 18, 2023.
- Built demo applications for Intel products and presented them at in-person AI PC workshops.
- Collaborated with Guy Tamir on Intel's hands-on AI PC workshop at [GenML 2024](#), demonstrating local AI workloads on Intel hardware.
- Currently co-developing an accredited Practical AI course for Bar-Ilan University with Guy Tamir, translating current AI tools and research into hands-on course material.
- Authored example code used in Intel's official [YouTube AI tutorials](#).

PubNub Technical Content Collaboration

- Contributed live technical guidance, code, and research explanations to PubNub CTO Stephen Blum's programming and AI livestreams since 2025.
- Wrote a hands-on PyTorch guide and prepared code and pull requests used on stream to explain CNNs, NLP embeddings, and click-prediction models.
- Received a written recommendation from [Stephen Blum, PubNub co-founder and CTO](#).

Selected Projects

- [datalog_par](#): A standards-oriented Datalog compiler and query engine written in Rust, using magic sets and Rayon for parallel evaluation.
- [source_viewer](#): A language-agnostic Rust TUI for exploring disassembly and source mappings from DWARF debug information.
- [frame_mem_utils](#): Stack-backed Rust arenas and vector-like data structures for working without direct heap allocation.
- [auto_new](#): A lightweight, dependency-free Rust procedural macro that generates constructors while avoiding unnecessary stack copies.
- [GPU Cellular Automata](#): A Rust and wgpu simulation engine that runs cellular automata entirely on the GPU using WGS� shaders.

Language Design & Compiler Work

- Core team member of [PAL](#): A new programming language currently under closed development.
- Built two toy compilers/interpreters:
 - One in **pure C99 + NASM**, including a full optimization pipeline (constant folding, loop unrolling, branch elimination).
 - One in **Rust**, using `unsafe` for a simple dynamic language with a VM.

Open Source & Community Contributions

- Collaborate on API design and implementation for the Rust diagnostics library [Ariadne](#), across its earlier [GitHub repository](#) and current development on Codeberg.
- Helped contributors across several beginner-led programming-language projects through guidance, debugging, and occasional code contributions; also worked on an optimization for an open-source [B compiler](#).

Educational Content

- [Blog on Medium](#): Writings on systems programming.
- [benchmark-errors](#): Educational repo with curated examples of common benchmarking pitfalls and statistical analysis.

Side Projects

- **Movie Subtitles Translation Tool**: Built a [CLI pipeline](#) to batch-translate subtitle files, replacing hours of manual work for my mentor.
- **Accidental Rebuild of pypi-timemachine**: Recreated functionality for [reproducible PyPI builds](#).
- [vpn_proxy](#): Simple tool for managing VPN exit IPs; made as a weekend learning project.
- ...and more.

Technical Skills

AI & ML: PyTorch, TensorFlow, Hugging Face, Diffusers, OpenVINO, Intel IPEX

LLMs & Tooling: LangChain, OpenAI API, FAISS

Systems: Rust (including unsafe Rust and procedural macros), C, C++, x86-64 assembly (NASM, GAS), LLVM IR, CUDA (basic), Vulkan (basic), GLSL (basic), WGSL (basic), wgpu (basic)

Data & Scientific: NumPy, Pandas, scikit-learn, Matplotlib

Backend: Python, Rust, C, C++, PostgreSQL (basic)

Frontend & GUI (basic): JavaScript, HTML/CSS, egui, raylib, wgpu

Education

B.Sc. in Computer Science

- The Open University of Israel
- Currently in fourth year